

Financial Instrument Similarity — Statistical Features Cheat Sheet

Core principle: Similarity = similar behaviour + similar risk drivers + similar risk magnitude + similar economics.

$$R_A = \alpha + \beta R_B + \varepsilon \quad | \quad \varepsilon = \text{actual A return} - \text{predicted A return}$$

Feature	What it tells you	Similar instruments should have	Priority
Correlation	Do they move together?	High correlation	High
Beta	How strongly A reacts to B	Appropriate, stable beta	High
Volatility	Typical movement magnitude	Similar volatility	High
Volatility ratio	Relative risk scale: Vol(A)/Vol(B)	Near 1	High
Residual mean	Average unexplained difference	Near 0	Medium
Residual volatility	Typical unexplained divergence	Low	Very high
Max residual	Largest divergence	Small	High
Residual skewness	Asymmetric divergence	Similar / acceptable	Medium
Residual kurtosis	Extreme divergence / tails	Similar / controlled	Medium-high
Residual autocorrelation	Whether divergence persists	Depends on purpose	Medium
Rolling correlation	Stability of co-movement	High and stable	Very high
Rolling beta	Stability of sensitivity	Stable	Very high
Rolling residual vol	Whether relationship breaks	Low/stable	Very high
Stress residual	Divergence during shocks	Small	Very high
Factor exposure	Why they move together	Similar factor betas	Very high

Residual Analysis — The Similarity Engine

Residuals measure what is left after one asset explains another. Small, stable residuals generally indicate a tighter relationship.

- 1. Residual mean:** ϵ . Close to zero is desirable, but positive and negative errors can cancel.
- 2. Residual volatility:** $SD(\epsilon)$. Often the strongest simple residual-based measure. Lower = less unexplained movement.
- 3. Maximum residual:** $\max|\epsilon|$. Finds occasional large divergences hidden by averages.
- 4. Skewness:** identifies asymmetric divergence; useful for credit/derivatives.
- 5. Kurtosis:** identifies fat-tail divergence and extreme events.
- 6. Autocorrelation:** persistent residuals can indicate missing factors, liquidity, maturity effects or mean reversion.
- 7. Rolling residual volatility:** checks whether similarity is stable across regimes.
- 8. Stress residual:** measure divergence during equity sell-offs, rate shocks, credit widening, FX stress or high-volatility periods.

Residual pattern	Interpretation
Small random residuals around 0	Strong relationship; little unexplained difference
Large residual volatility	Weak similarity or missing risk factors
Residual trend	Relationship may be changing / model misspecification
Residual clusters	Volatility regime or liquidity effects
Large residual spikes	Tail divergence / poor substitution in stress
Persistent residuals	Possible maturity, liquidity, factor or mean-reversion effect

Asset-Type Statistical Fingerprints

Asset type	Statistical features	Risk / economic features	Residual focus
Equity	Return corr, beta, volatility, downside beta, rolling corr/beta	Market, sector, size, value, momentum, quality/growth factors	Residual vol, stress residual, stability
Equity derivatives	P&L;/return corr, vol, rolling beta/corr	Delta, gamma, vega, theta, rho, IV, moneyness, expiry	Residual P&L; after Greeks/factor model
Bonds	Yield-change corr, return corr, spread/yield vol, rolling beta	Duration, DV01, convexity, key-rate duration, spread, rating, seniority, maturity	Residual yield/spread change; tenor substitution
CDS	Spread-change corr, spread vol, beta, rolling measures	CS01, spread, maturity, rating, sector, country, recovery, hazard/default risk	Residual spread change; stress widening
FX spot	Return corr, beta, volatility, rolling corr/beta	Rate differential, carry, inflation, central-bank/commodity sensitivity	Residual FX return; stress behaviour
FX options	P&L;/return corr and volatility behaviour	IV, skew, risk reversal, term structure, Greeks	Residual after delta/vega/factor adjustment

Rating: for bonds/CDS, treat rating as an economic constraint or ordinal distance, not as an ordinary continuous variable. Combine it with issuer/sector, seniority, currency and maturity.

Tenor substitution: compare [Duration, DV01, Convexity, Spread, Rating] rather than maturity alone. A 6Y BBB bond can be a better substitute for a 5Y BBB target than a 10Y BBB bond if its risk fingerprint is closer.

Production Similarity Workflow

Layer	Check	Action
1. Economic filters	Asset class, currency, underlying, issuer/sector, rating, seniority, maturity, liquidity	Reject clearly incompatible candidates
2. Market behaviour	Correlation, beta, volatility, volatility ratio, downside correlation	Rank behavioural similarity
3. Residual behaviour	Mean, residual vol, max residual, skew, kurtosis, autocorrelation	Measure unexplained difference
4. Stability	Rolling correlation/beta/residual volatility	Detect changing relationships
5. Stress	Stress correlation, beta, residual	Test substitution when markets are stressed
6. Risk factors	Equity factors; bond duration/DV01; CDS CS01; option Greeks; FX drivers	Confirm economic similarity
7. Long-run relationship	Cointegration, stationarity, mean reversion where appropriate	Useful for relative-value relationships
8. Overall score	Weighted distances / similarity score	Create ranked candidate list

Distance = w_{return} ·return-distance + w_{risk} ·risk-distance + w_{factor} ·factor-distance + $w_{residual}$ ·residual-distance + $w_{economic}$ ·economic-distance

Similarity = $\exp(-Distance)$

Practical priority: 1) risk-factor compatibility → 2) residual volatility → 3) stress residual → 4) correlation → 5) beta → 6) volatility ratio → 7) rating compatibility → 8) cointegration → 9) residual mean.

GOLDEN RULE

High correlation ≠ identical risk. Strong similarity means high co-movement + similar risk magnitude/sensitivity + small, stable residuals + compatible economics.